JUAN SEBASTIÁN SÁNCHEZ URBANO

Data Analyst · Python, SQL and Power BI · Industrial Automation Engineer

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LinkedIn · GitHub · Portfolio

PROFESSIONAL PROFILE

Data analyst with an industrial automation engineering background. I turn plant, sales and operations data into indicators and dashboards that support production and commercial decisions. I work with SQL, Python, R and Power BI, and design medallion-architecture (Bronze/Silver/Gold) ETL pipelines.

TECHNICAL SKILLS

Data analysis: Advanced SQL, MySQL, Python, R, Advanced Excel, Power BI, Dashboards, Statistical analysis, EDA, Data consolidation.

Machine Learning & AI: Machine Learning, Predictive modeling, Classification, AI agents, AI chatbots.

Data engineering: ETL pipelines, Medallion architecture, Data quality and cleaning, REST APIs, AWS.

Sales & industrial automation: Technical and commercial advisory, Sales analysis, n8n, Industrial automation, PLC programming, CNC and robotics, Real-time IoT sensors.

Web development & deployment: HTML, CSS, JavaScript, Bash, YAML, Git and GitHub, GitHub Actions, GitHub Pages.

Methodology & Management: PDCA (continuous improvement), Analytical problem solving, Results orientation, Effective communication.

PROFESSIONAL EXPERIENCE

Technical, Sales and Data Automation Advisor — Proeléctricos · Popayán, Cauca

2019 – Nov 2025 · Mar 2026 – Sep 2026

- I analyze and validate customers' technical and operational information to structure comprehensive solutions, ensuring the quality and traceability of commercial records.
- **Achievement:** Achievement: Designed an n8n pipeline that automatically extracts, validates and routes customer reports and information to the responsible teams, with a web chatbot.

CNC and Robotics Area Lead (Production Data Analysis) — Metálicas Mundial · Candelaria, Valle del Cauca

Oct 2025 – Jan 2026

- I led the capture and analysis of critical production and quality-control variables, ensuring the reliability of manufacturing process data.
- **Achievement:** I identified bottlenecks and optimized cycle times by analyzing machinery operating data under agile continuous-improvement methodologies, increasing the area's productivity.

FEATURED PROJECTS

Predictive maintenance with Machine Learning

Simulated data · Machine Learning, Classification, Python

- Fault-severity classification model built from machine signals, with cross-validation.
- **Impact:** Cross-validation accuracy of 74% on 720 simulated signals, performing best on high-severity faults.

Production dashboard: OEE and Pareto

Simulated data · Python, Dashboards, OEE, Pareto

- OEE dashboard (availability, performance and quality) and Pareto analysis of stoppages to prioritize improvements on the production line.
- **Impact:** Example with simulated data: reference OEE of 70.8%, with 92.3% availability and 94.0% quality.

SIVIGILA epidemiological ETL 2018–2022

Dengue · SARI · Tuberculosis · Python, Excel, Medallion, HTML dashboard

- An ETL process over five National Institute of Health Excel files, with extraction, transformation and loading scripts, medallion architecture and an HTML dashboard to visualize results.

EDUCATION

Complementary training

- Artificial Intelligence Bootcamp (MinTIC / Talento Tech) · Data Science, Python and Machine Learning (Platzi) Platzi profile

Industrial Automation Engineering

Universidad del Cauca · Graduated 2025